

Curcumin in various cancers.

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Curcumin (diferuloylmethane), an active constituent of turmeric, is a well-described phytochemical, which has been used since ancient times for the treatment of various diseases. The dysregulation of cell signaling pathways by the gradual alteration of regulatory proteins is the root cause of cancers. Curcumin modulates regulatory proteins through various molecular mechanisms. Several research studies have provided in-depth analysis of multiple targets through which curcumin induces protective effects against cancers including gastrointestinal, genitourinary, gynecological, hematological, pulmonary, thymic, brain, breast, and bone. The molecular mechanisms of action of curcumin in treating different types of cancers remain under investigation. The multifaceted role of this dietary agent is mediated through its inhibition of several cell signaling pathways at multiple levels. Curcumin has the ability to inhibit carcinogenicity through the modulation of the cell cycle by binding directly and indirectly to molecular targets including transcription factors (NF- κ B, STAT3, β -catenin, and AP-1), growth factors (EGF, PDGF, and VEGF), enzymes (COX-2, iNOS, and MMPs), kinases (cyclin D1, CDKs, Akt, PKC, and AMPK), inflammatory cytokines (TNF, MCP, IL-1, and IL-6), upregulation of proapoptotic (Bax, Bad, and Bak) and downregulation of antiapoptotic proteins (Bcl(2) and Bcl-xL). A variety of animal models and human studies have proven that curcumin is safe and well tolerated even at very high doses. This study elaborates the current understanding of the chemopreventive effects of curcumin through its multiple molecular pathways and highlights its therapeutic value in the treatment and prevention of a wide range of cancers. Copyright © 2013 International Union of Biochemistry and Molecular Biology, Inc.